sky_map

April 4, 2023

[ ]:
```
# import libraries
import math
import random
import matplotlib.pyplot as plt
```

[ ]:
```
# generate random coordinates for 9000 stars
long = []    # array to store longitudes
lat = []     # array to store latitudes

for i in range (9000):
    # generate numbers between -pi and pi
    long.append (random.random() * (2 * math.pi) - math.pi)

    # generate numbers between -pi/2 and pi/2
    lat.append (random.random() * (math.pi) - (math.pi)/2)
```

[ ]:
```
# create the plot
plt.figure (figsize = (16, 8.4))
plt.subplot (projection = "aitoff")
plt.title ("Map of the Sky")
plt.plot (long, lat, 'o', markersize = 1, alpha = 0.4, color = 'r')
plt.grid (True)
plt.savefig ('sky_map.pdf')
plt.show()
```